Notebook 1 – Chapter 1.1

Combinatorics & Discrete Counting Foundations

20 worked examples for the Jane Street National Probability Challenge, several adapted from Mosteller's *Fifty Challenging Problems in Probability*

A. Fundamental Counting Rules

Example 1 – Sum Rule vs. Product Rule

(a) A desk can route an order through 5 exchanges *or* 3 dark pools (never both). How many venue choices? (b) Independently it also picks 1 of 4 order types and 1 of 3 sizes. How many full configurations (venue, type, size)?

Solution. (a) Disjoint choices ⇒ sum rule: $5 + 3 = 8$. (b) Independent successive choices ⇒ product rule: $8 \times 4 \times 3 = 96$.

Example 2 – Combinations via a Perfect Deal (Mosteller Prob. 8)

From a well-shuffled 52-card deck dealt 13 cards, find $P(\text{all 13 are the same suit})$.

Solution. Order doesn't matter, so count with combinations: only $\binom{13}{13} = 1$ way to pick all of one suit, times 4 suits. Total 13-card hands: $\binom{52}{13}$. Hence

$$P = \frac{4}{\binom{52}{13}} = \frac{4 \cdot 13!\,39!}{52!} \approx 6.3 \times 10^{-12}.$$

(Mosteller derives the same number via raw permutations $52 \cdot 51 \cdots 40$; dividing by 13! to kill ordering recovers $\binom{52}{13}$ – this is exactly why $\binom{n}{k} = P(n,k)/k!$.)

Example 3 – Pascal's Rule, Combinatorial Proof

Prove $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Solution. Count k-subsets of $\{1, \ldots, n\}$ by whether element n is included. If included, choose the remaining $k-1$ from the other $n-1$: $\binom{n-1}{k-1}$ ways. If excluded, choose all k from the other $n-1$: $\binom{n-1}{k}$ ways. These cases partition all k-subsets, so they sum to $\binom{n}{k}$.

Example 4 – Binomial Sum Identity

Show $\sum_{k=0}^{n} \binom{n}{k} = 2^n$, and note why $\sum_k \binom{n}{k} p^k q^{n-k} = 1$.

Solution. Each subset of an n-set is obtained by an independent in/out choice per element: $2 \cdot 2 \cdots 2 = 2^n$ subsets total, and subsets of size k number $\binom{n}{k}$; summing over k counts every subset once. Setting $p+q=1$ in the Binomial Theorem $(p+q)^n = \sum \binom{n}{k} p^k q^{n-k}$ gives $1^n = 1$ – the reason $\text{Bin}(n,p)$ probabilities always sum to 1.

Example 5 – Circular Permutations

8 traders sit around a round table for poker. How many distinct seatings (a) up to rotation, (b) up to rotation *and* reflection?

Solution. (a) Fix one trader to kill rotational symmetry: the rest arrange in $(8-1)! = 7! = 5040$ ways. (b) Reflections identify each arrangement with its mirror image, halving the count (for $n \geq 3$): $5040/2 = 2520$.

B. Stars and Bars

Example 6 – Basic Stars and Bars

Split 10 identical capital units among 4 distinct strategies, any strategy may get 0. How many allocations?

Solution. Nonnegative integer solutions to $x_1 + x_2 + x_3 + x_4 = 10$: $\binom{10+4-1}{4-1} = \binom{13}{3} = 286$.

Example 7 – Lower-Bound Constraint

Same as Ex. 6, but every strategy must receive *at least* 1 unit.

Solution. Substitute $y_i = x_i - 1 \geq 0$: $\sum y_i = 10 - 4 = 6$. Count $= \binom{6+4-1}{4-1} = \binom{9}{3} = 84$.

Example 8 – Capacity Cap via Complementation

Same setup as Ex. 6, but a position limit caps strategy 1 at ≤ 3 units.

Solution. Total unrestricted (Ex. 6) = 286. Subtract violators $x_1 \geq 4$: set $x_1' = x_1 - 4 \geq 0$, so $x_1' + x_2 + x_3 + x_4 = 6$, giving $\binom{6+3}{3} = \binom{9}{3} = 84$ violators. Valid count $= 286 - 84 = 202$.

Example 9 – Inequality via Slack Variable

Count nonnegative integer solutions of $x_1 + x_2 + x_3 \leq 7$ (idle lots allowed across 3 books).

Solution. Introduce slack $x_4 \geq 0$ so $x_1+x_2+x_3+x_4 = 7$: $\binom{7+4-1}{4-1} = \binom{10}{3} = 120$.

Example 10 – Capped Stars and Bars: Dice Sums (cf. Craps, Prob. 9)

In how many ways can three distinguishable dice sum to 10?

Solution. Solve $x_1 + x_2 + x_3 = 10$, $1 \leq x_i \leq 6$. Shift $y_i = x_i - 1$: $\sum y_i = 7$, $0 \leq y_i \leq 5$. Unrestricted: $\binom{7+2}{2} = 36$. Subtract cases $y_i \geq 6$ (set $y_1' = y_1 - 6$, remaining sum = 1, $\binom{1+2}{2} = 3$ ways, times 3 choices of which y_i) $= 9$; no two variables can simultaneously exceed 5 since $6 + 6 > 7$. Answer: $36 - 9 = 27$ ways – matches the classical count of 27 rolls totaling 10 with three dice.

C. Inclusion–Exclusion & Derangements

Example 11 – PIE with Three Sets

How many integers from 1 to 1000 are divisible by 3, 5, or 7?

Solution. With $\lfloor \cdot \rfloor$ counts: $|A| = 333, |B| = 200, |C| = 142$; pairwise $|A \cap B| = 66, |A \cap C| = 47, |B \cap C| = 28$; triple $|A \cap B \cap C| = 9$.

$$|A \cup B \cup C| = 333 + 200 + 142 - 66 - 47 - 28 + 9 = 543.$$

Example 12 – Derangement Probability, Direct

10 people each bring a wrapped gift; gifts are redistributed by a uniform random permutation. Find $P(\text{no one gets their own gift back})$.

Solution.

$$P = \frac{D_{10}}{10!}, \qquad D_n = n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}.$$

$D_{10} = 1{,}334{,}961$, $10! = 3{,}628{,}800$, so $P \approx 0.367879 \approx 1/e$.

Example 13 – Derangement Recurrence

Given $D_3 = 2$, $D_4 = 9$, use $D_n = (n-1)(D_{n-1} + D_{n-2})$ to find D_5, and verify against the series formula.

Solution. $D_5 = 4(D_4 + D_3) = 4(9 + 2) = 44$. Check: $5!\left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120}\right) = 120(0.3\overline{6}) = 44$. ✓

Example 14 – Exactly r Fixed Points (matches Mosteller's Matching-Problem formula)

Of 6 shuffled, re-dealt cards, in how many permutations do exactly 2 land in their original position?

Solution. Choose the 2 fixed positions, then derange the rest:

$$\binom{6}{2} D_4 = 15 \times 9 = 135.$$

Dividing by 6! reproduces Mosteller's own matching-problem formula (Prob. 46) $P(r \mid n) = \frac{1}{r!} P(0 \mid n-r)$, since $\binom{n}{r} D_{n-r}/n! = D_{n-r}/\big(r!(n-r)!\big)$.

Example 15 – Rapid Convergence to $1/e$ (Mosteller Probs. 45–46)

For the 52-card matching game (compare two shuffled decks card-by-card), estimate $P(0 \text{ matches})$ and compare to the $n \to \infty$ limit.
Solution. $P(0 \mid 52) = D_{52}/52! = \sum_{i=0}^{52}(-1)^i/i! \approx 0.3678794412$, indistinguishable from $e^{-1} = 0.3678794412\ldots$ to 9 decimal places – the alternating factorial series converges essentially by $n = 10$, which is why Mosteller's Poisson approximation for matching/derangement counts is already excellent for modest n.

D. Catalan Numbers, Compositions & Partitions

Example 16 – Ballot-Type Path Counting (cf. The Ballot Box, Prob. 22)

Two candidates each get $n = 3$ votes; ballots are drawn one at a time. How many drawing sequences keep candidate A weakly ahead (tally$_A \geq$ tally$_B$) at every point? (This lattice-path count is the Catalan number C_n.)
Solution. $C_n = \dfrac{1}{n+1}\binom{2n}{n}$. For $n = 3$: $C_3 = \frac{1}{4}\binom{6}{3} = \frac{1}{4}(20) = 5$.
(Contrast with Mosteller's strict-lead / tie-probability computation in Prob. 22, which sums over the *complementary* lattice paths.)

Example 17 – Dyck Paths / Balanced Parentheses

How many valid sequences of 4 buy/sell pairs keep cumulative sells $\leq$ cumulative buys at every prefix (equivalently, 4 balanced parentheses pairs)?
Solution. $C_4 = \frac{1}{5}\binom{8}{4} = \frac{1}{5}(70) = 14$.

Example 18 – Non-Crossing Matchings

10 simultaneous order arrivals sit on a circle; pair them into 5 buy/sell matches with no crossing lines. How many non-crossing pairings exist?
Solution. Non-crossing perfect matchings of $2n$ points $= C_n$. For $n = 5$: $C_5 = \frac{1}{6}\binom{10}{5} = \frac{1}{6}(252) = 42$.

Example 19 – Compositions: Ordered Dice Sums

How many ways can 3 distinguishable dice show values summing to 12?
Solution. Solve $x_1 + x_2 + x_3 = 12$, $1 \leq x_i \leq 6$; shift $y_i = x_i - 1$: $\sum y_i = 9$, $0 \leq y_i \leq 5$. Unrestricted: $\binom{9+2}{2} = 55$. Subtract $y_i \geq 6$: set $y_1' = y_1 - 6$, remaining sum $= 3$, $\binom{3+2}{2} = 10$, times $3 = 30$ (two variables ≥ 6 simultaneously is impossible since $12 > 9$). Answer: $55 - 30 = 25$ – the well-known count of 25 ways to roll a 12 with three dice.

Example 20 – Compositions vs. Partitions of the Same Sum

For a target sum of 10: (a) how many *ordered* triples of positive integers sum to 10 (compositions)? (b) how many *unordered* multisets of at most 3 positive parts sum to 10 (partitions)?
Solution. (a) Compositions of 10 into exactly 3 positive parts: $\binom{10-1}{3-1} = \binom{9}{2} = 36$. (b) Partitions of 10 into at most 3 parts: listing by number of parts, 1 part: $\{10\}$ (1 way); 2 parts: $(9,1), (8,2), (7,3), (6,4), (5,5)$ (5 ways); 3 parts: $(8,1,1), (7,2,1), (6,3,1), (6,2,2), (5,4,1), (5,3,2), (4,4,2), (4,3,3)$ (8 ways). Total $= 1 + 5 + 8 = 14$. The gap between 36 and 14 is exactly the overcounting that "order matters" introduces – the key trap in dice/card sum-enumeration problems.

Drill instruction: re-derive each boxed result from scratch (no peeking) before your next Stage-1 timed set. Flag any derivation you cannot reconstruct in under 90 seconds – that is your weak topic for the next session.